Graded linear algebra:

signs, everywhere

Let us start with basic definitions and results concerning graded vector spaces, algebras, manifolds etc.

1 Graded vector spaces

Recall that a $\mathbb{Z}$-graded real vector space V can be defined as follows.

Definition 1.1 [$\mathbb{Z}$-graded vector space]**.** A $\mathbb{Z}$-graded vector space V is a vector space which admits a direct sum decomposition of the form

$$V = \bigoplus_{k\in\mathbb{Z}} V_k \,, \tag{1.1}$$

where each V_k is a vector space, called the subspace of homogeneous elements of degree k, or homogeneous subspace of degree k.

A morphism of $\mathbb{Z}$-graded vector spaces is a linear map $f : V \longrightarrow W$ which preserves the grading, i.e. which $f(V_k) \subseteq W_k$ for all $k \in \mathbb{Z}$.

If a graded vector space V has only homogeneous subspaces of degrees $k \in \mathbb{N}$, it will be called $\mathbb{N}$-graded. If instead V has only non-trivial homogeneous subspaces of degrees $k \in \{n, n+1, \dots, m\}$ for some integers $n \leq m$, then it will be said to be concentrated in degrees n to m. A common example of graded vector space is that of a simple Lie algebra, as it always admits a decomposition of the form

$$\mathfrak{g} = \mathfrak{n}_+ \oplus \mathfrak{h} \oplus \mathfrak{n}_+ \,, \tag{1.2}$$

where $\mathfrak{h}$ is a Cartan subalgebra, and $\mathfrak{n}_\pm$ are both (maximal) nilpotent subalgebras, generated by the raising and lowering operators respectively. As the notation suggests, it is quite natural to assign degree 0 to the Cartan subalgebra, and degree ± 1 to the nilpotent subalgebras $\mathfrak{n}_\pm$. The paradigmatic example is the angular momentum algebra $\mathfrak{so}(3) = \mathrm{span}\{J_a\}$, whose Cartan subalgebra can be chosen as the one generated by J_3, while the ladder operators $J_\pm$ generate the nilpotent subalgebras $\mathfrak{n}_\pm$ respectively.

The subalgebras $\mathfrak{h} \Subset \mathfrak{n}_\pm$ are called the *standard Borel subalgebras* of $\mathfrak{g}$, which are maximal solvable subalgebras. An interesting generalisation of these subalgebras are *parabolic subalgebras* of $\mathfrak{g}$, which can be defined as any subalgebras of $\mathfrak{g}$ which contains a Borel subalgebra. A choice of parabolic subalgebras induces a grading on $\mathfrak{g}$ of the form

$$\mathfrak{g} = \mathfrak{p}_- \oplus \mathfrak{g}_0 \oplus \mathfrak{p}_+ \,, \qquad \mathfrak{p}_\pm = \bigoplus_{k=1}^{N} \mathfrak{g}_{\pm k} \,, \tag{1.3}$$

where $\mathfrak{g}_0$ is a reductive algebra (direct sum of semisimple Lie algebra and a center), $\mathfrak{p}_\pm$ are nilpotent and the parabolic subalgebra is $\mathfrak{p} = \mathfrak{g}_0 \Subset \mathfrak{p}_+$. In fact, gradations of these types (concentrated in degrees $-N$ to N) are in bijecive correspondence with parabolic subalgebras (see [1] for a textbook account).

An example commonly encountered in field theory is that of the conformal algebra in n-dimensions, $\mathfrak{g} = \mathfrak{so}(2, n)$, and its parabolic subalgebra $\mathfrak{g}_0 \Subset \mathfrak{g}_{+1}$ where $\mathfrak{g}_0 = \mathfrak{so}(1,1) \oplus \mathfrak{so}(n)$ is generated by dilations and rotations, and $\mathfrak{g}_{+1} = \mathbb{R}^n$ by special conformal transformations.

Finally, another standard example of graded vector space is the that of differential forms on a manifold Σ,

$$\Omega_\Sigma = \bigoplus_{k=0}^{\dim\Sigma} \Omega^k_\Sigma \,, \tag{1.4}$$

where the gradation is nothing but the form degree, and hence Ω_Σ is concentrated in degrees $0, \dots, \dim\Sigma$.

(∗) Remark 1.1. Note that the existence of a direct sum of homogeneous subspaces is not necessary: we could simply define $\mathbb{Z}$-graded vector spaces as a collection $\{V_k\}_{k\in\mathbb{Z}}$ of vector spaces indexed by integers. This would prevent us from considering inhomogeneous elements, but in many contexts it will not be a problem—considering homogeneous elements would be sufficient. In fact, such a relaxed definition is even desirable in order for sheaves of graded algebras to be defined correctly, as explained in the recent paper [2].

(∗) Remark 1.2. Note that one could replace $\mathbb{Z}$ with any other Abelian group G, and define in a similar manner G-graded vector spaces, their tensor or symmetric algebras and their suspension (see e.g. [3, Chap. 21]).

Definition 1.2 [Suspension]. The p-suspension of a $\mathbb{Z}$-graded vector space V is the graded vector space denoted by $V[p]$, and whose homogeneous subspaces are defined as

$$V[p]_k := V_{p+k}\,, \qquad k\in\mathbb{Z}\,, \tag{1.5}$$

i.e. they are those of V but with a shift in degree.

In plain words, the suspension operation is simply a uniform shift of degree of all homogeneous subspaces: elements which were of degree k in V (i.e. which belong to the subspace V_k) are now considered as elements of degree $k-p$ in $V[p]$ (since, reversing the above relation, we have $V_k = V[p]_{k-p}$).

This operation appears in many places in the literature, and can also come in handy when dealing with linear maps with an intrinsic degree. Indeed, one could easily consider a map between two graded vector spaces $f : V \longrightarrow W$ such that $f(V_k) \subseteq W_{k+p}$, i.e. that sends elements in V of degree k to elements in W of degree $k+p$. We will write the space of such maps as

$$\mathrm{Hom}^p_{\mathbb{R}}(V,W) := \left\{ f : V \longrightarrow W \mid f(V_k) \subseteq W_{k+p}\,,\ \forall k\in\mathbb{Z} \right\}, \tag{1.6}$$

where the subscript $\mathbb{R}$ refers to the fact that these are $\mathbb{R}$-*linear* maps, and the superscript p indicates their degree.

So far there is not much to the grading introduced on vector spaces, except that it gives us a convenient way to single out particular subspaces. On its own, this can already be quite useful, as can be seen in the examples taken from the theory of simple Lie algebras. But the gradation becomes a more important ingredient when one starts considering vector spaces with additional structure, such a product (to be discussed next), or a differential.

Indeed, given a $\mathbb{Z}$-graded vector space V, one can consider a linear map $\delta_V : V \to V$ on itself of *degree* 1,

$$\ldots \longrightarrow V_{-1} \xrightarrow{\delta_V} V_0 \xrightarrow{\delta_V} V_{+1} \longrightarrow \ldots\,, \tag{1.7}$$

and that squares to 0, i.e. $\delta_V^2 = 0$. In other words, (V,δ_V) is a $\mathbb{Z}$-graded vector space endowed with a differential — sometimes called a differential graded vector space or *dg-vector space* — and hence defines a cochain complex of vector spaces.

Since we have added more structure, it is only natural to require morphisms between dg-vector spaces to preserve it. Here, this would mean considering degree-preserving linear maps $f : V \longrightarrow W$ which *intertwine* the differentials, i.e.

$$\delta_W f(v) = f(\delta_V v)\,, \qquad \forall v\in V\,. \tag{1.8}$$

Note that one can also consider a linear map with a non-zero intrinsic degree between V and W, in which case the condition that it intertwines the differential reads

$$\delta_W f(v) = (-1)^{|f|}\, f(\delta_V v)\,, \qquad \forall v\in V\,, \tag{1.9}$$

so that we see a first sign (pun intended) of the grading of V and W. In fact, we will see shortly that most of the subtlety in dealing with graded objects has to do with signs.

Before moving on, let us introduce a complex of maps between two complexes of vector spaces.

Definition 1.3 [Complex of graded linear maps]**.** Given two complexes of vector spaces (V, δ_V) and (W, δ_W), the space of linear maps from V to W of *any* degree,

$$\mathrm{Hom}_{\mathbb{R}}(V, W) = \bigoplus_{k\in\mathbb{Z}} \mathrm{Hom}^k_{\mathbb{R}}(V, W)\,, \tag{1.10}$$

is a complex with differential $\partial_{V,W}$ defined as the commutator of the two differentials, i.e.

$$\partial_{V,W} f = \delta_W \circ f - (-1)^{|f|}\, f \circ \delta_V\,, \tag{1.11}$$

for any map $f \in \mathrm{Hom}^{|f|}_{\mathbb{R}}(V, W)$.

This complex allows one to encode, amongst other things, morphisms of complexes as cocycles of degree 0. Indeed, the condition $\partial_{V,W} f = 0$ implies that the linear map $f : V \longrightarrow W$ *intertwines* the differential of V and W, which is what is required of a morphism of complexes. Similarly, *homotopies* between such morphisms can be encoded in terms of this complex.

2 Tensor, symmetric and exterior algebra

So far, we have seen that graded vector spaces are simply vector spaces with an additional 'decoration', its decomposition into a direct of homogeneous subspaces. Unsurprisingly, most—if not all—operations defined on vector spaces also carry on to graded vector space, starting with the direct sum and tensor product.

Definition 2.1 [Direct sum and tensor product]**.** Given two $\mathbb{Z}$-graded vector spaces V and W, their direct sum $V \oplus W$, and their tensor product $V \otimes W$ are also $\mathbb{Z}$-graded vector spaces, with homogeneous subspaces of degree $k \in \mathbb{Z}$ defined as

$$(V \oplus W)_k := V_k \oplus W_k\,, \qquad (V \otimes W)_k := \bigoplus_{n\in\mathbb{Z}} V_n \otimes V_{k-n}\,, \tag{2.1}$$

where one the right hand side of both equations, the direct sum and tensor product symbols should be understood as those of vector spaces.

The tensor algebra of any vector space V is the graded associative algebra denoted by $T(V)$ whose underlying vector space is defined as the direct sum of all tensor products of V,

$$T(V) := \bigoplus_{n=0}^{\infty} V^{\otimes n}\,, \tag{2.2}$$

where $V^{\otimes 0} = \mathbb{R}$. The associative product is simply the concatenation of tensors,

$$V^{\otimes k} \otimes V^{\otimes l} \longrightarrow V^{\otimes k+l}\,, \tag{2.3}$$

which extends to $T(V)$ by linearity. Its associativity follows from the associativity of the tensor product of vector spaces.

Concretely, given a basis $\{\mathsf{t}_a\}$ of V, a tensor $\tau \in T(V)$ take the form,

$$\tau = \tau^{a_1\dots a_n}\, \mathsf{t}_{a_1} \otimes \cdots \otimes \mathsf{t}_{a_n}\,, \tag{2.4}$$

where the components $\tau^{a_1\dots a_n}$ do *not have* any symmetries in their indices. The product of two such tensors is simply given by

$$\begin{aligned} \tau_1 \cdot \tau_2 &= \big(\tau_1^{a_1\dots a_m}\, \mathsf{t}_{a_1} \otimes \cdots \otimes \mathsf{t}_{a_m}\big) \otimes \big(\tau_2^{b_1\dots b_n}\, \mathsf{t}_{b_1} \otimes \cdots \otimes \mathsf{t}_{b_n}\big) \\ &= \tau_1^{a_1\dots a_m}\, \tau_2^{b_1\dots b_n}\, \mathsf{t}_{a_1} \otimes \cdots \otimes \mathsf{t}_{a_m} \otimes \mathsf{t}_{b_1} \otimes \cdots \otimes \mathsf{t}_{b_n}\,, \end{aligned} \tag{2.5}$$

i.e. it defines a new tensor whose components are obtained by the product of the components of the two tensors. Note however that the *order* of the components matter here, in the sense that $\tau_1 \cdot \tau_2$ has components $\tau_1^{a_1\dots a_m}\, \tau_2^{a_{m+1}\dots a_{m+n}}$, whereas the components of $\tau_2 \cdot \tau_1$ are $\tau_2^{a_1\dots a_n}\, \tau_1^{a_{n+1}\dots a_{m+n}}$, which are *not* the same.

Since no symmetry is imposed on tensors here, one finds no distinction between the cases where V is graded or not. So let us imposing some anti/symmetry properties to tensors on V.

Definition 2.2 [Symmetric and exterior algebra]**.** The symmetric and exterior algebras of a $\mathbb{Z}$-graded vector space are respectively defined as the quotient

$$S(V) := T(V)/\mathcal{I}_+ \,, \qquad \text{and} \qquad \wedge V := T(V)/\mathcal{I}_- \,, \tag{2.6}$$

where $\mathcal{I}_\pm$ is the ideal

$$\mathcal{I}_\pm := \left\langle v \otimes w \pm (-1)^{|v||w|}\, w \otimes v \,|\, \forall\, v, w \in V \right\rangle , \tag{2.7}$$

i.e. generated by the identification of tensors related by the action of the symmetric group.

More explicitly, a basis for symmetric tensors on V of rank 2 is given by

$$\mathsf{t}_a \odot \mathsf{t}_b = (-1)^{|a||b|}\, \mathsf{t}_b \odot \mathsf{t}_a \,, \tag{2.8}$$

whereas for rank 3, it reads

$$\mathsf{t}_a \odot \mathsf{t}_b \odot \mathsf{t}_c = (-1)^{|a||b|}\, \mathsf{t}_b \odot \mathsf{t}_a \odot \mathsf{t}_c = (-1)^{|a|(|b|+|c|)}\, \mathsf{t}_b \odot \mathsf{t}_c \odot \mathsf{t}_a = \dots \,, \tag{2.9}$$

and so on. More generally, the permutation of elements in a product produces a sign,

$$v_1 \odot \cdots \odot v_n = \varepsilon(\sigma; v_1, \dots, v_n)\, v_{\sigma(1)} \odot \cdots \odot v_{\sigma(n)} \,, \tag{2.10}$$

that is often called the *Koszul sign*, and which corresponds to the sign obtained by re-arranging the elements v_k using the rule that when v_i passes through v_j, it produces $(-1)^{|v_i||v_j|}$.

One can easily check that if V is concentrated in degree 0, then we have the following commutative algebra isomorphism,

$$S(V[1]) \cong \wedge V \tag{2.11}$$

i.e. the symmetric algebra of the suspension of V is nothing but the exterior algebra of V itself.

(∗) Remark 2.1. Note that this isomorphism holds *only* when V is concentrated in degree 0. For a general $\mathbb{Z}$-graded vector space V, one has the isomorphism of $\mathbb{Z}$-graded vector spaces

$$\mathrm{dec} : S^n(V[1]) \xrightarrow{\sim} (\wedge^n V)[n] \,, \tag{2.12}$$

often called the '*décalage* isomorphism' (from the French words for 'shift'). This isomorphism takes the form

$$\mathrm{dec}\big(s(v_1) \odot \cdots \odot s(v_n)\big) = (-1)^{\sum_{k=1}^n (n-k)(|v_k|+1)}\, s^n\big(v_1 \wedge \cdots \wedge v_n\big) \,, \tag{2.13}$$

where

$$s : V \longrightarrow V[1] \,, \tag{2.14}$$

sends a element $v \in V$ to itself in the suspension $V[1]$, and $s^n : V \longrightarrow V[n]$ the composition of this suspension isomorphism n times. The sign ensures the compatibility of the graded commutative and anti-commutative structures on both sides. However, this is *not* an algebra morphism.

3 Differential graded Lie algebras

One can also consider Lie brackets on graded vector spaces, and as in the case of commutative associative products, ask that this Lie bracket is compatible with the grading, leading to the notion of (differential) graded Lie algebra.

Definition 3.1 [Differential graded Lie algebra]**.** A differential graded Lie algebra (dg-Lie algebra) is a graded vector space $\mathcal{L}$, endowed with

- a differential $\partial : \mathcal{L} \longrightarrow \mathcal{L}$, i.e. a nilpotent linear operator of degree 1 on $\mathcal{L}$, so that $(\mathcal{L}, \partial)$ is a complex;
- a graded Lie bracket

  $$[-,-] : \mathcal{L} \wedge \mathcal{L} \longrightarrow \mathcal{L} \,, \tag{3.1}$$

  meaning it is a bilinear operation, antisymmetric in the $\mathbb{Z}$-graded sense,

  $$[x,y] = -(-1)^{|x||y|}\, [y,x] \,, \qquad \forall x, y \in \mathcal{L} \,, \tag{3.2}$$

  and the $\mathbb{Z}$-graded Jacobi identity,

  $$[x,[y,z]] = [[x,y],z] + (-1)^{|x||y|}\, [y,[x,z]] \,, \qquad \forall x, y, z \in \mathcal{L} \,, \tag{3.3}$$

  so that $(\mathcal{L}, [-,-])$ a $\mathbb{Z}$-graded Lie algebra;

and such that the differential is a derivation for the Lie bracket, i.e. it obeys the Leibniz rule

$$\partial\,[x,y] = [\partial x, y] + (-1)^{|x|}\,[x,\partial y]\,, \tag{3.4}$$

for any elements $x, y \in \mathcal{L}$.

The most common example of dg-Lie algebra, at least in field theory, is that the algebra of differential forms on a manifold Σ valued in a Lie algebra $\mathfrak{g}$, i.e. $\Omega_\Sigma \otimes \mathfrak{g}$. Taking $\mathfrak{g}$ to be concentrated in degree 0, the total degree is simply the form degree, so that elements of degree 1 are $\mathfrak{g}$-valued 1-forms, $A \in \Omega^1_\Sigma \otimes \mathfrak{g}$ — Yang–Mills gauge fields on Σ — while elements of degree 0 are $\mathfrak{g}$-valued 0-forms, $\varepsilon \in \Omega^0_\Sigma \otimes \mathfrak{g}$ — the associated gauge parameters.

In fact, this example is an instance of the following construction: given a dgca $(\mathcal{A}, \mathrm{d}_\mathcal{A}, \wedge)$ and a dg-Lie algebra $(\mathcal{L}, \partial_\mathcal{L}, [-,-]_\mathcal{L})$, one can define a structure of differential graded Lie algebra on the tensor product of $\mathcal{A}$ and $\mathcal{L}$, namely

$$\big(\mathcal{A}\otimes\mathcal{L}, \partial_{\mathcal{A}\otimes\mathcal{L}}, [-,-]_{\mathcal{A}\otimes\mathcal{L}}\big) \tag{3.5}$$

is a dg-Lie algebra, with the differential defined as

$$\partial_{\mathcal{A}\otimes\mathcal{L}}(a\otimes x) = \mathrm{d}_\mathcal{A} a \otimes x + (-1)^{|a|}\, a \otimes \partial_\mathcal{L} x\,, \tag{3.6}$$

while the Lie bracket is defined as

$$[a\otimes x, b\otimes y]_{\mathcal{A}\otimes\mathcal{L}} = (-1)^{|x||b|}\,(a\wedge b)\otimes [x,y]_\mathcal{L}\,, \tag{3.7}$$

for any element $a, b \in \mathcal{A}$ and $x, y \in \mathcal{L}$. We will leave the verification that the above does satisfy the defining conditions of a dg-Lie algebra (nilpotency of $\partial_{\mathcal{A}\otimes\mathcal{L}}$, Leibniz rule and Jacobi identity) to the interested reader.

(∗) Remark 3.1. Let us mention that the formula for the differential on $\mathcal{A}\otimes\mathcal{L}$ can be written, without evaluating it on elements, as

$$\partial_{\mathcal{A}\otimes\mathcal{L}} = \mathrm{d}_\mathcal{A}\otimes \mathbf{1}_\mathcal{L} + \mathbf{1}_\mathcal{A}\otimes\partial_\mathcal{L}\,, \tag{3.8}$$

which, upon using the Koszul sign rule yields (3.6). Similarly, the Leibniz rule for the Lie bracket of a dg-Lie algebra reads,

$$\partial_\mathcal{L}\,[-,-]_\mathcal{L} = [\partial_\mathcal{L}(-),-]_\mathcal{L} + [-,\partial_\mathcal{L}(-)]_\mathcal{L}\,, \tag{3.9}$$

which reproduces (3.4) upon using the Koszul sign rule again.

The Maurer–Cartan elements of the dgla $(\mathcal{L}, \partial_\mathcal{L}, [-,-]_\mathcal{L})$ are elements $x \in \mathcal{L}_1$ of degree 1 which obey

$$\partial_\mathcal{L} x + \tfrac{1}{2}\,[x,x]_\mathcal{L} = 0\,, \tag{3.10}$$

known as the Maurer–Cartan equation. The space of Maurer–Cartan elements, often denoted by

$$\mathsf{MC}(\mathcal{L}) := \big\{x\in\mathcal{L}_1 \mid \partial_\mathcal{L} x + \tfrac{1}{2}\,[x,x]_\mathcal{L} = 0\big\}\,, \tag{3.11}$$

is acted upon by the Lie group that integrates the degree 0 Lie subalgebra $\mathcal{L}_0$.

References

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